AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

Smart bank insights:AI Powered Banking analytics using Power BI and Copliot

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Project Domain

Power BI with Copilot

A Project Report

Submitted in partial fulfilment of the requirements of the AICW Project

by:

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Year - 2026

ACKNOWLEDGEMENT

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Firstly, I would like to thank our supervisor and mentor Ms Umamaheswari for being a great mentor and the best adviser we could have asked for. Her advice, encouragement, and constructive feedback served as a source of innovative ideas and inspiration, and were central to the successful completion of this project.

I would also like sincerely thank Edunet Foundation, Microsoft, LinkedIn and SAP for providinding me with this valuable opportunity.The knowledge about AI and practical exposure offered through this platform have significantly enhenced my skills into thw world of AI.

I am are also grateful to my institution,Graphic Era hill university for providing the necessary infrastructure and support throughout the project duration.

Finally,Thanks to my Parents and Friends for motivating in best possible way.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

The project titled Smart bank insights:AI Powered Banking analytics using Power BI and Copilot ”is based on using data and AI tools to understand banking information in a simple and effective way. In today’s time, banks deal with a large amount of data, and it is important to analyze this data to make better decisions and improve services.

In this project, Power BI is used to create easy-to-understand dashboards and visual reports that show important details like customer transactions, loan data, and overall performance. Along with this, Microsoft Copilot helps in generating quick insights, summaries, and suggestions, making the analysis faster and easier.

The main aim of this project is to help banks understand their data clearly, identify patterns, and improve their services. By using AI and data visualization together, this project shows how technology can make banking analysis more efficient and user-friendly.

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CHAPTER 1

Introduction

1.1 Problem Statement

The core problem that my project addresses.

A.)What is the main problem or challenge in your chosen domain?

ans. Below are the some problems that I have faced will using MS excel with copilot as domain

Difficulty handling large dataset

Data security issusse

Limited adv.analytical capabilities

Human errors

B.)Why is this problem significant, and what is its real-world impact?

Ans. Banks generate large amount of financial and customer data every day thus,Managing the data only with Excel can be difficult especially when the data set is very large Sometimes Excel files may contain errors such as duplicates missing value on incorrect formulas which can affect the accuracy of the banking record even though manual data handling In that case the risk of the human error and slow possessing.

The real world impact of the problem can may impact Into the wrong Financial decision and errors in data can affect loan approval data customer records In the financial record of the bank Which may reduce The efficiency and reliability of the banking operation

C.)Who is affected by this problem — what are the target users or organizations?

Ans. The target user or organization which is further affected by the problem are Bank managers, Bank employees, Data analyst, It and operation team, customers and government

D.)What are the limitations of existing approaches or tools currently available?

Ans. The below limitations Shows that while Excel with copilot Improve the productivity But it is still required A human supervision An additional tools for the advanced analysis So here are the some existing Limitation Of Excel with Microsoft copilot.

Limited Advanced Analytics

Possibility of incorrect AI suggestions

Security concerns

Dependency data quality

Limited handling of very large data set

1.2 Motivation

I choose banking and financial data for this project Because I come from a commerce background where the understanding of Financial Banking Information is important as well as very crucial The banking sector unearths a large amount of data which needs a proper analysis for the better decision making .If we look at in the today’s scenario All kind of sectors present in our economy

Including the banking sector AI has become Revelation technology In many organization Are using AI tools for that analysis Therefore This project use Modern tools Like Excel and copilot To analyse banking data more efficiently **.**

1.3 Objectives

1. To collect and understand banking and financial data used in the project.
2. To clean and organizethe datasets by removing duplicates, correcting formats, and handling missing values.
3. To apply domain-specific analysis using Excel with Copilot.
4. To build interactive dashboards or reports that clearly communicate insights.
5. To leverage AI tools (Copilot / Chatbot) to automate or enhance the analysis and content generation process.
6. To identify patterns in trend in banking data

1.4 Scope of the Project

Define the boundaries and limitations of the project clearly.

- This project covers data from [describe the dataset — e.g., sales records, patient records, employee payroll, etc.].The data set is about banking customer dataset itcolumns names username,riskcatrogry, KYC satuts etc.

- The tools used are limited to [Power BI / Copilot] as assigned.

- The analysis focuses on - Data cleaning, removing unwanted columns (i.e further not reverence in analysis)in power query editor and making a efficient dashboard for analysis.

- Monitor Key performance indicator(KPIs) account status,risk category etc.

- Improve accuracy and efficiency data interpretation by using an interactive power BI dashboard.

CHAPTER 2

Literature Survey

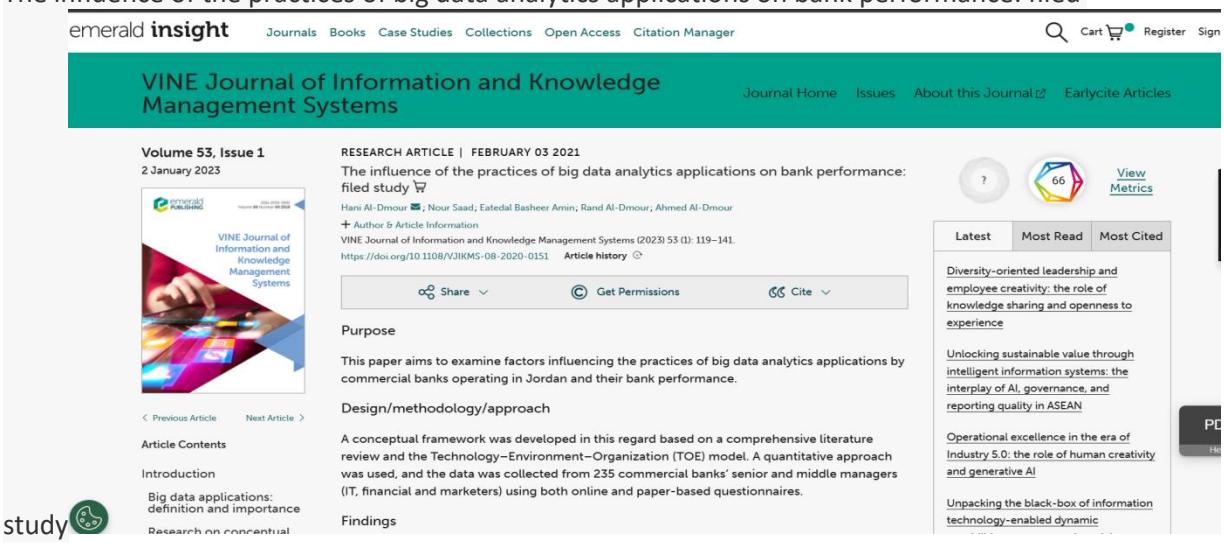
A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

- Review relevant literature or previous work in my project domain

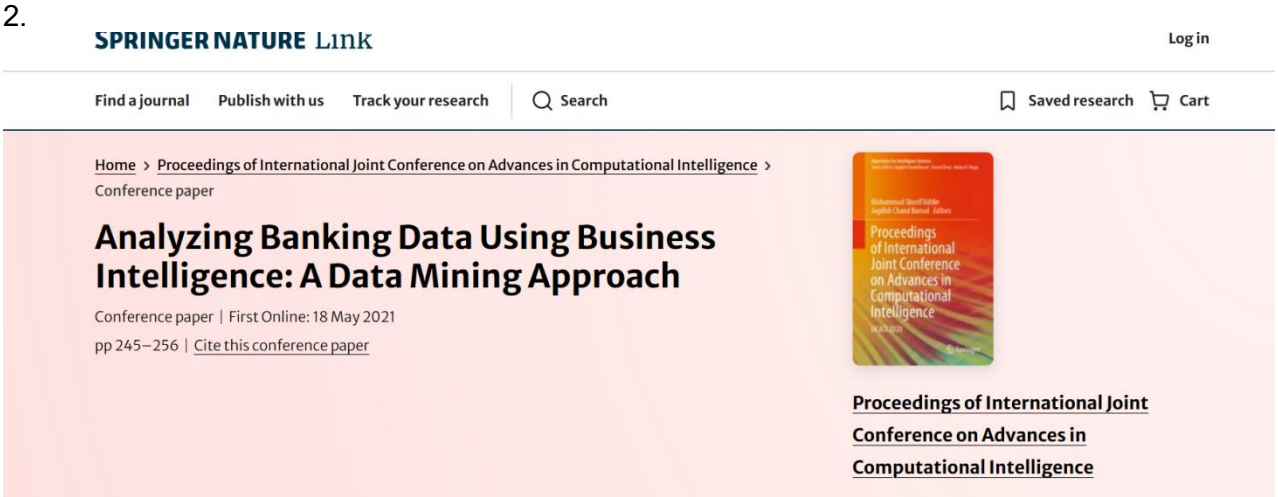
1. RESEARCH ARTICLE ON FEBRUARY 03 2021

The influence of the practices of big data analytics applications on bank performance: filed study



The influence of the practices of big data analytics applications on bank performance: filed study | VINE Journal of Information and Knowledge Management Systems | Emerald Publishing

2.



Analyzing Banking Data Using Business Intelligence: A Data Mining Approach | Springer Nature Link

2.2 Tools and Technologies Review

Briefly review the tools and technologies that are used in my project:

- Microsoft Power BI — Microsoft Power BI is a powerful data visualization and business analytics tool used to convert raw data into meaningful insights. It allows users to create interactive dashboards, charts, and reports that make data easy to understand. In this project, Power BI is used to analyze banking data such as customer transactions, loan details, and performance trends. It helps in identifying patterns, tracking key metrics, and supporting better decision-making. Its user-friendly interface makes it suitable even for beginners to explore and present data effectively.

- Microsoft Copilot (AI) —Microsoft Copilot is an AI-powered tool that helps users analyze data and generate insights quickly and easily. It works alongside tools like Power BI to provide automatic summaries, suggestions, and answers based on the data. In this project, Copilot helps in understanding banking data by highlighting key trends and patterns without requiring complex analysis. It reduces manual effort and makes the overall process faster and more efficient, even for users with basic technical knowledge.

2.3 Gaps Identified

- Most existing literature focuses on complex machine learning approaches; simpler, tool-based solutions for non-technical users are underexplored.
- There is limited documentation on integrating AI Copilot with Excel/Power BI for domain-specific analysis in the context of student projects.
- Prompt engineering best practices for business use cases are still evolving and lack standardized frameworks.
- Existing research focuses more on data collection, but less on practical implementation using tools like Power BI.
- Limited attention on small-scale or student-level projects that demonstrate real-world application.
- Gap in providing quick and automated summaries for faster decision-making.
- Most research does not fully explore how AI tools can reduce manual effort in analysis.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

Describe the overall design approach of your project. Explain how data flows from input to output in your system.

- Data Source / Input:Kaggledata set that is contains 1500 rows
- Data Cleaning: Power BI query editor remove unwanted columns which further has mo relevance in analysis(username, user-id, phone no.)
- Analysis / Processing: By using Power BI visualization tools like graphs chats etc
- Visualization / Output: The final output — Interactive dashboard of banking customer data.

3.2 Modules / Components Used

List the key modules or functional components of your project:

- Module 1 — Data Ingestion: Loading the raw dataset into Power BI.
- Module 2 — Data Cleaning: Removing inconsistencies, duplicates, and null values and the data which has no relevance in further analysis(username, phone no.,user-id)
- Module 3 — Data Analysis: using various charts to data analysis.
- Module 4 — Visualization / Report Generation: Building dashboardsin power BI
- Module 5 — AI Integration: Using Copilot for insight summaries.

3.3 Advantages

List the key advantages of your proposed approach:

- Automates repetitive data cleaning tasks, saving significant manual effort.
- Provides clear, visual insights that are easy to understand for non-technical stakeholders.
- Leverages AI tools (Copilot) to accelerate analysis and reduce human error.
- Scalable — the same methodology can be applied to larger datasets or different domains.
- Cost-effective — uses widely available tools (Power BI) that are part of the Microsoft 365 ecosystem.
- User friendly,so even non- technical users can easily use it.
- Saves time by atomating data analyisis tasks.
- Helps in better decision making using accurate data.

3.4 Requirement Specification

Hardware Requirements

Component	Specification

Processor	Intel Core i5
RAM	16 GB
Storage	50 GB
Display	1920x 1080 resolution
Internet	Required for Copilot

Software Requirements

Software	Details
Operating System	Windows 11
Microsoft Excel	Microsoft 365 (Excel with Copilot)
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	No
AI Chatbot	Microsoft Copilot (web)
Web Browser	Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

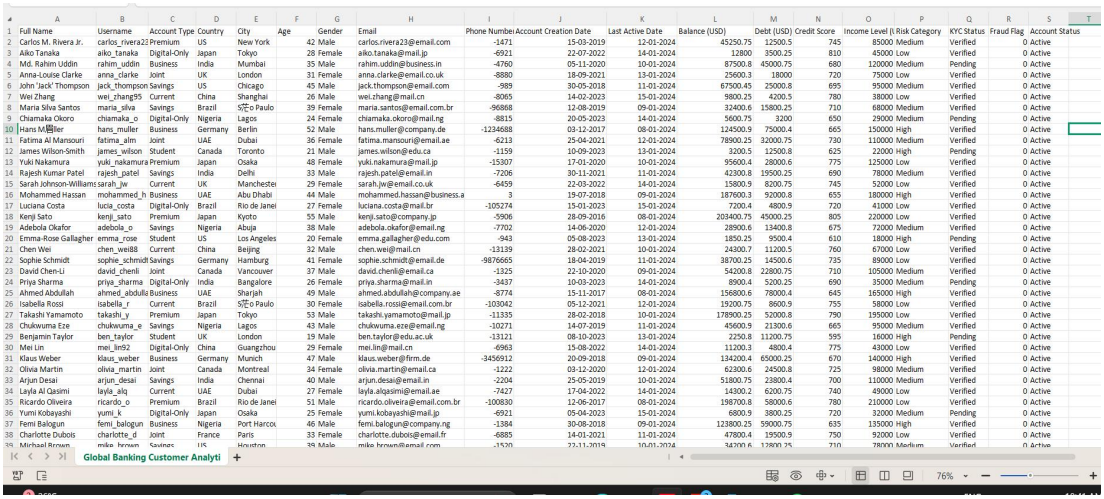
This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Describe the data cleaning process and present the before/after comparison of your dataset.

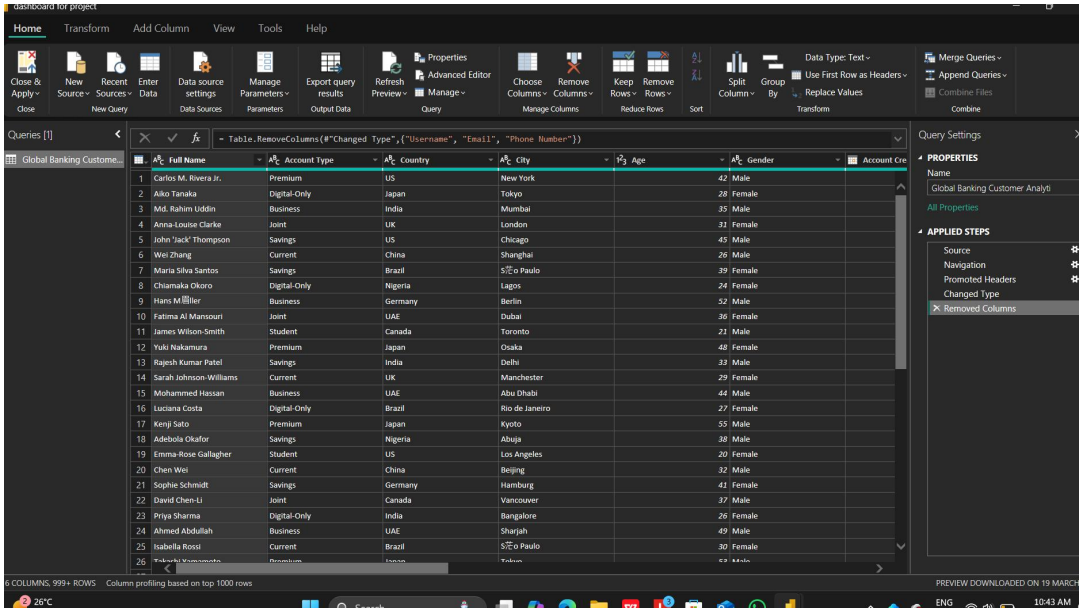
- Dataset source: Downloaded from Kaggle
- Original dataset size: originally 19 columns,1599 rows were there in dataset
- Issues found: Non relevant data columns in the data set and some duplicates
- Cleaning steps applied: Power BI (Transform data to power query editor then remove columns then remove duplicates)
- Cleaned dataset size: after final cleaning there were 16 columns and 1500 rows.

Uncleaned data sets



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Full Name	Username	Account Type	Country	City	Age	Gender	Email	Phone Numbe	Account Creation Date	Last Active Date	Balance (USD)	Debt (USD)	Credit Score	Income Level (\	Risk Category	KYC Status	Fraud Flag	Account Status
2	Carlos M. Rivera Jr.	carlos_rivera23	Premium	US	New York	42	Male	carlos.rivera23@email.com	-1471	15-03-2019	12-01-2024	45250.75	12500.5	745	85000	Medium	Verified	0	Active
3	Aiko Tanaka	aiko_tanaka	Digital-Only	Japan	Tokyo	28	Female	aiko.tanaka@mail.jp	-6921	22-07-2022	14-01-2024	12800	3500.25	810	45000	Low	Verified	0	Active
4	Md. Rahim Uddin	rahim_uddin	Business	India	Mumbai	35	Male	rahim.uddin@business.in	-4760	05-11-2020	10-01-2024	87500.8	45000.75	680	120000	Medium	Pending	0	Active
5	Anna-Louise Clarke	anna_clarke	Joint	UK	London	31	Female	anna.clarke@email.co.uk	-8880	18-09-2021	13-01-2024	25600.3	18000	720	75000	Low	Verified	0	Active
6	John 'Jack' Thompson	jack_thompson	Savings	US	Chicago	45	Male	jack.thompson@email.com	-989	30-05-2018	11-01-2024	67500.45	25000.8	695	95000	Medium	Verified	0	Active
7	Wei Zhang	wei_zhang95	Current	China	Shanghai	26	Male	wei.zhang@mail.cn	-8065	14-02-2023	15-01-2024	9800.25	4200.5	780	38000	Low	Verified	0	Active
8	Maria Silva Santos	maria_silva	Savings	Brazil	São Paulo	39	Female	maria.santos@email.com.br	-96868	12-08-2019	09-01-2024	32400.6	15800.25	710	68000	Medium	Verified	0	Active
9	Chiamaka Okoro	chiamaka_o	Digital-Only	Nigeria	Lagos	24	Female	chiamaka.okoro@mail.ng	-8815	20-05-2023	14-01-2024	5600.75	3200	650	29000	Medium	Pending	0	Active
10	Hans Müller	hans_muller	Business	Germany	Berlin	52	Male	hans.muller@company.de	-1234688	03-12-2017	08-01-2024	124500.9	75000.4	665	150000	High	Verified	0	Active
11	Fatima Al Mansouri	fatima_alm	Joint	UAE	Dubai	36	Female	fatima.mansouri@email.ae	-6213	25-04-2021	12-01-2024	78900.25	32000.75	730	110000	Medium	Verified	0	Active
12	James Wilson-Smith	james_wilson	Student	Canada	Toronto	21	Male	james.wilson@edu.ca	-1159	10-09-2023	13-01-2024	3200.5	12500.8	625	22000	High	Pending	0	Active
13	Yuki Nakamura	yuki_nakamura	Premium	Japan	Osaka	48	Female	yuki.nakamura@mail.jp	-15307	17-01-2020	10-01-2024	95600.4	28000.6	775	125000	Low	Verified	0	Active
14	Rajesh Kumar Patel	rajesh_patel	Savings	India	Delhi	33	Male	rajesh.patel@email.in	-7206	30-11-2021	11-01-2024	42300.8	19500.25	690	78000	Medium	Verified	0	Active
15	Sarah Johnson-Williams	sarah_jw	Current	UK	Manchester	29	Female	sarah.jw@email.co.uk	-6459	22-03-2022	14-01-2024	15800.9	8200.75	745	52000	Low	Verified	0	Active
16	Mohammed Hassan	mohammed_h	Business	UAE	Abu Dhabi	44	Male	mohammed.hassan@business.a	3	19-07-2018	09-01-2024	187600.3	92000.8	655	180000	High	Verified	0	Active
17	Luciana Costa	lucia_costa	Digital-Only	Brazil	Rio de Janei	27	Female	luciana.costa@mail.br	-105274	15-01-2023	15-01-2024	7200.4	4800.9	720	41000	Low	Verified	0	Active
18	Kenji Sato	kenji_sato	Premium	Japan	Kyoto	55	Male	kenji.sato@company.jp	-5906	28-09-2016	08-01-2024	203400.75	45000.25	805	220000	Low	Verified	0	Active
19	Adebola Okafor	adebola_o	Savings	Nigeria	Abuja	38	Male	adebola.okafor@email.ng	-7702	14-06-2020	12-01-2024	28900.6	13400.8	675	72000	Medium	Verified	0	Active
20	Emma-Rose Gallagher	emma_rose	Student	US	Los Angeles	20	Female	emma.gallagher@edu.com	-943	05-08-2023	13-01-2024	1850.25	9500.4	610	18000	High	Pending	0	Active
21	Chen Wei	chen_wei88	Current	China	Beijing	32	Male	chen.wei@mail.cn	-13139	28-02-2021	10-01-2024	24300.7	11200.5	760	67000	Low	Verified	0	Active
22	Sophie Schmidt	sophie_schmidt	Savings	Germany	Hamburg	41	Female	sophie.schmidt@email.de	-9876665	18-04-2019	11-01-2024	38700.25	14500.6	735	89000	Low	Verified	0	Active
23	David Chen-Li	david_chenli	Joint	Canada	Vancouver	37	Male	david.chenli@email.ca	-1325	22-10-2020	09-01-2024	54200.8	22800.75	710	105000	Medium	Verified	0	Active
24	Priya Sharma	priya_sharma	Digital-Only	India	Bangalore	26	Female	priya.sharma@mail.in	-3437	10-03-2023	14-01-2024	8900.4	5200.25	690	35000	Medium	Pending	0	Active
25	Ahmed Abdullah	ahmed_abdulla	Business	UAE	Sharjah	49	Male	ahmed.abdullah@company.ae	-8774	15-11-2017	08-01-2024	156800.6	78000.4	645	165000	High	Verified	0	Active
26	Isabella Rossi	isabella_r	Current	Brazil	São Paulo	30	Female	isabella.rossi@email.com.br	-103042	05-12-2021	12-01-2024	19200.75	8600.9	755	58000	Low	Verified	0	Active
27	Takashi Yamamoto	takashi_y	Premium	Japan	Tokyo	53	Male	takashi.yamamoto@mail.jp	-11335	28-02-2018	10-01-2024	178900.25	52000.8	790	195000	Low	Verified	0	Active
28	Chukwuma Eze	chukwuma_e	Savings	Nigeria	Lagos	43	Male	chukwuma.eze@email.ng	-10271	14-07-2019	11-01-2024	45600.9	21300.6	665	95000	Medium	Verified	0	Active
29	Benjamin Taylor	ben_taylor	Student	UK	London	19	Male	ben.taylor@edu.ac.uk	-13121	08-10-2023	13-01-2024	2250.8	11200.75	595	16000	High	Pending	0	Active
30	Mei Lin	mei_lin92	Digital-Only	China	Guangzhou	29	Female	mei.lin@mail.cn	-6963	15-08-2022	14-01-2024	11200.3	4800.4	775	43000	Low	Verified	0	Active
31	Klaus Weber	klaus_weber	Business	Germany	Munich	47	Male	klaus.weber@firm.de	-3456912	20-09-2018	09-01-2024	134200.4	65000.25	670	140000	High	Verified	0	Active
32	Olivia Martin	olivia_martin	Joint	Canada	Montreal	34	Female	olivia.martin@email.ca	-1222	03-12-2020	12-01-2024	62300.6	24500.8	725	98000	Medium	Verified	0	Active
33	Arjun Desai	arjun_desai	Savings	India	Chennai	40	Male	arjun.desai@email.in	-2204	25-05-2019	10-01-2024	51800.75	23800.4	700	110000	Medium	Verified	0	Active
34	Layla Al Qasimi	layla_alq	Current	UAE	Dubai	27	Female	layla.alqasimi@email.ae	-7427	17-04-2022	14-01-2024	14300.2	6200.75	740	49000	Low	Verified	0	Active
35	Ricardo Oliveira	ricardo_o	Premium	Brazil	Rio de Janei	51	Male	ricardo.oliveira@email.com.br	-100830	12-06-2017	08-01-2024	198700.8	58000.6	780	210000	Low	Verified	0	Active
36	Yumi Kobayashi	yumi_k	Digital-Only	Japan	Osaka	25	Female	yumi.kobayashi@mail.jp	-6921	05-04-2023	15-01-2024	6800.9	3800.25	720	32000	Medium	Pending	0	Active
37	Femi Balogun	femi_balogun	Business	Nigeria	Port Harcou	46	Male	femi.balogun@company.ng	-1384	30-08-2018	09-01-2024	123800.25	59000.75	635	135000	High	Verified	0	Active
38	Charlotte Dubois	charlotte_d	Joint	France	Paris	33	Female	charlotte.dubois@email.fr	-6885	14-01-2021	11-01-2024	47800.4	19500.9	750	92000	Low	Verified	0	Active
39	Michael Brown	mike_brown	Savings	US	Houston	38	Male	mike.brown@email.com	-1520	22-11-2019	10-01-2024	34300.6	13800.25	710	78000	Medium	Verified	0	Active

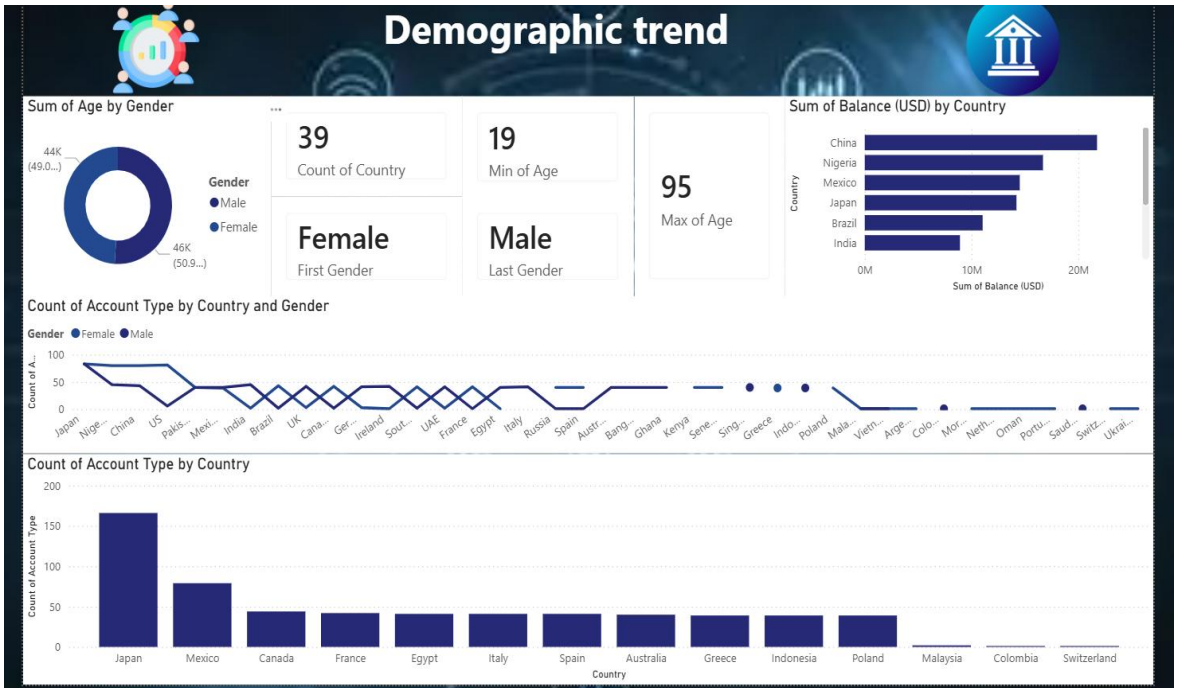
Cleaned data set



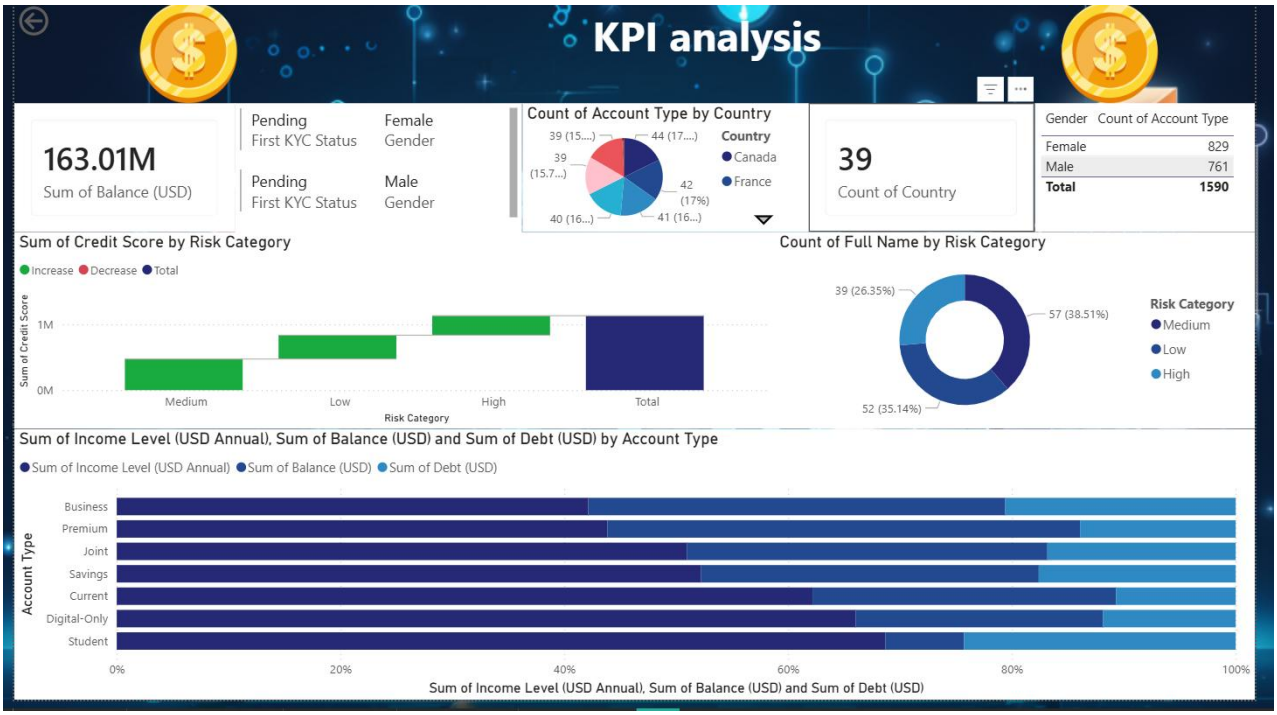
4.2 Visualization / Output Results

Present the key visualizations, dashboards, or generated content produced by your project.

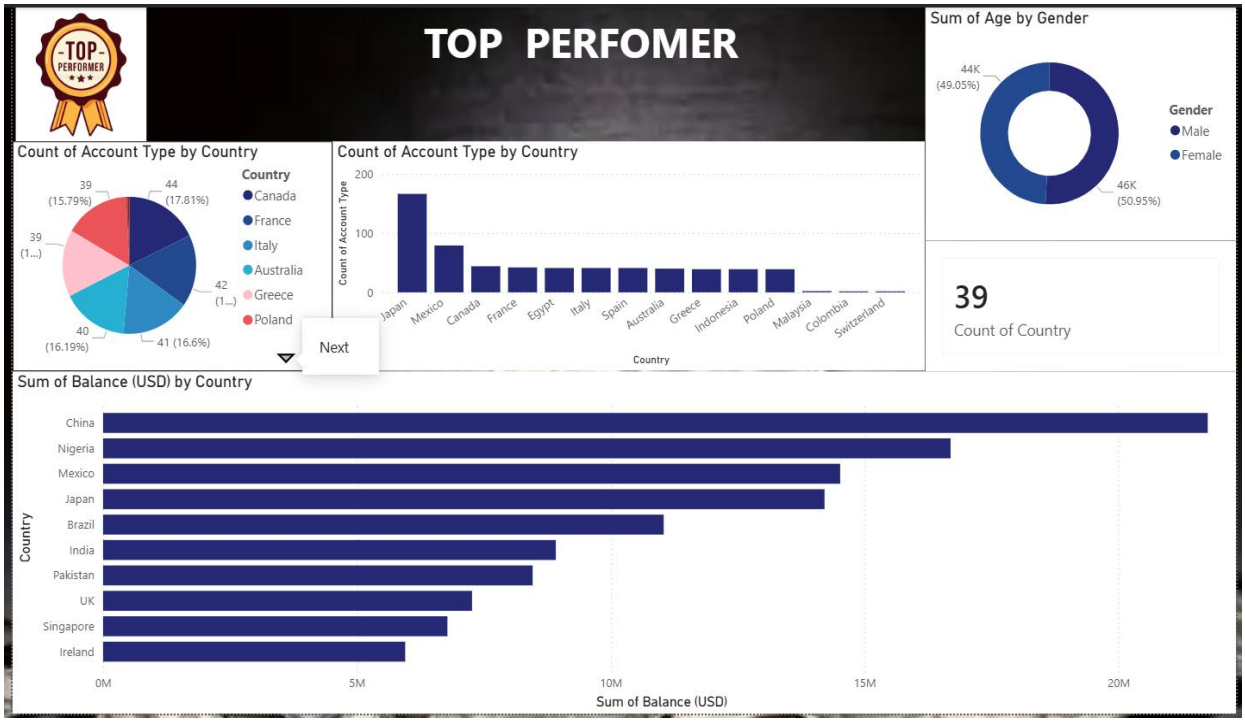
- Dashboard 1 :This visual is shows demographic trend in data



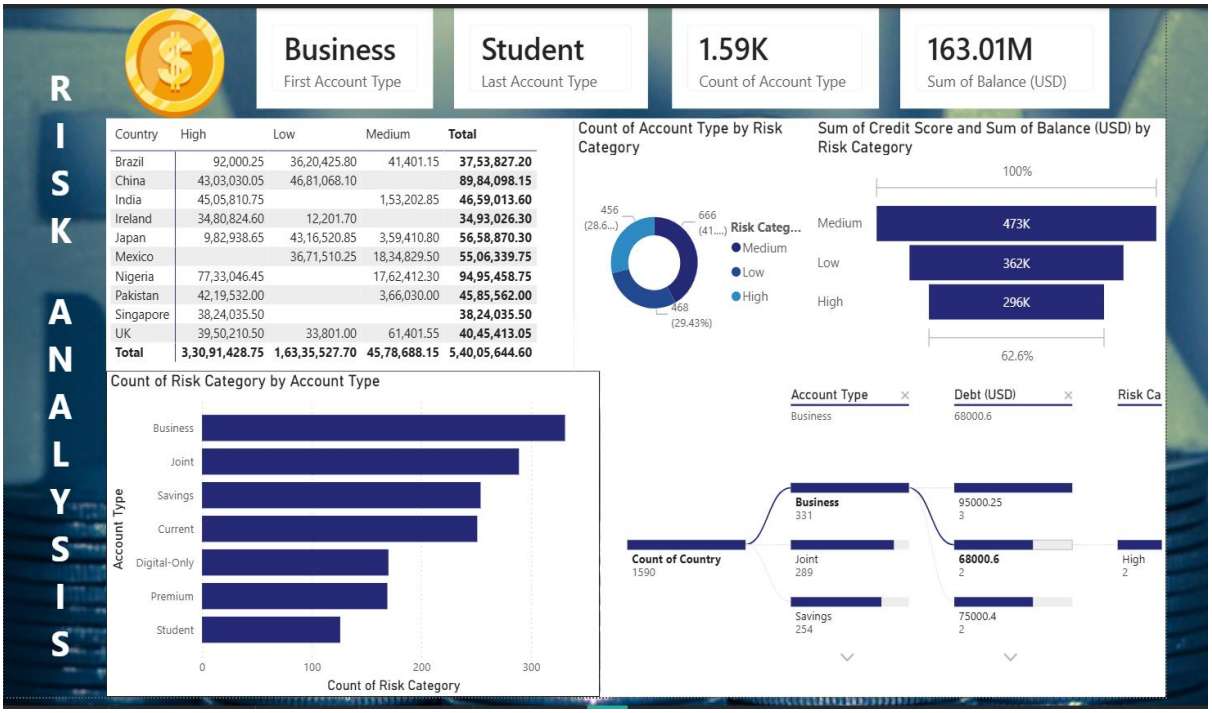
- Dashboard 2: This visual is of KPI analysis which shows key performance indicator



- Dashboard 3: This visual dashboard of top performer by various segment



- Dashboard4:



Country	High	Low	Medium	Total
Brazil	92,000.25	36,20,425.80	41,401.15	**37,53,827.20**
China	43,03,030.05	46,81,068.10		**89,84,098.15**
India	45,05,810.75		1,53,202.85	**46,59,013.60**
Ireland	34,80,824.60	12,201.70		**34,93,026.30**
Japan	9,82,938.65	43,16,520.85	3,59,410.80	**56,58,870.30**
Mexico		36,71,510.25	18,34,829.50	**55,06,339.75**
Nigeria	77,33,046.45		17,62,412.30	**94,95,458.75**
Pakistan	42,19,532.00		3,66,030.00	**45,85,562.00**
Singapore	38,24,035.50			**38,24,035.50**
UK	39,50,210.50	33,801.00	61,401.55	**40,45,413.05**
Total	**3,30,91,428.75**	**1,63,35,527.70**	**45,78,688.15**	**5,40,05,644.60**

4.3 AI Tool Integration Results

Describe how the AI tool (Copilot or Chatbot) was used in your project and what results it produced.

- Tool used: [Microsoft Copilot]

- Prompts used: **Prompt 1.)** Analyze the banking customer at a certain identify key financial tag such as average balance transaction pattern
- **Prompt 2.)** Provide the insight from the data set about with customer group health balance and better loan approval rates.

- Output quality:**Output of prompt 1.**)The copilot will generate the summary of the major financial patent data set helping to identify customer financial behaviour trends.
- **Output of prompt 2.**)Copilot identify customer segments with the stronger financial health based on their factor like income level, credit score, age.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Summarize the key results and insights discovered from the project.

- Key Finding 1: After cleaning, unwanted records and duplicate columns in data, I found that removing them significantly improved dashboard accuracy and performance.

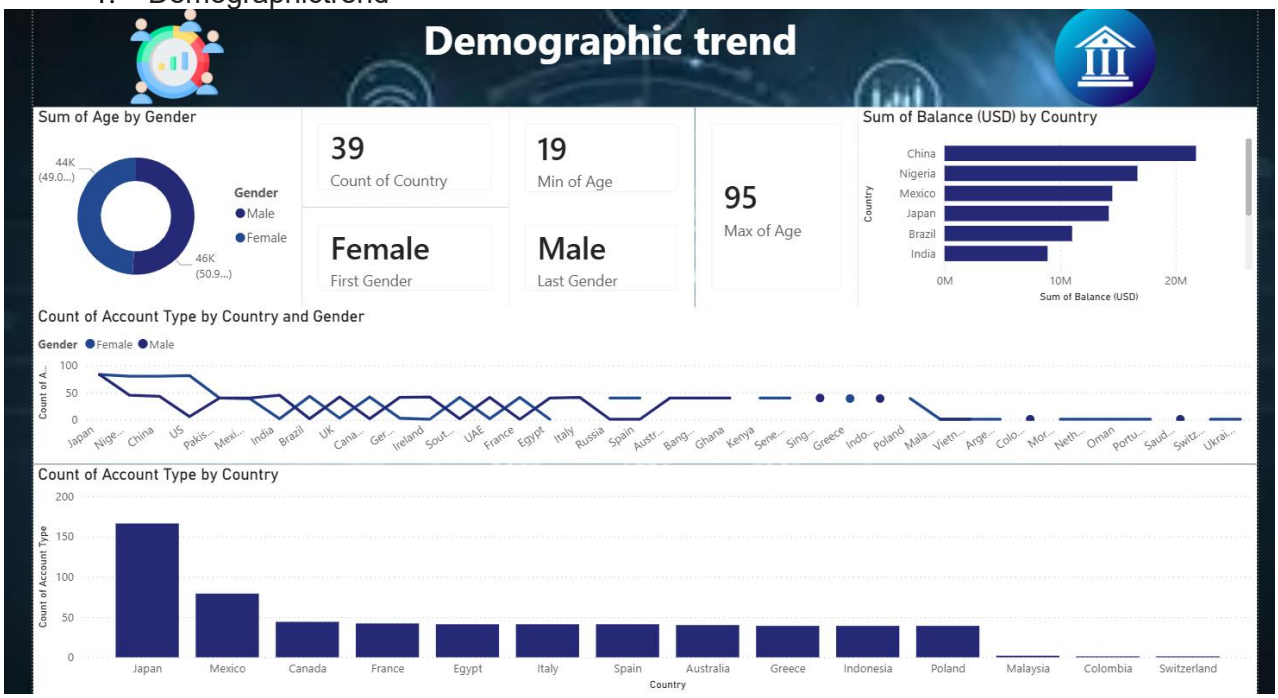
- Key Finding 2: The AI Copilot reduced the operational cost by minimizing manual work and resource use.

- Key Finding 3: The top 4 countries with maximum number of account holder are Japan, Nigeria, China and US which combines total 48% of global customers

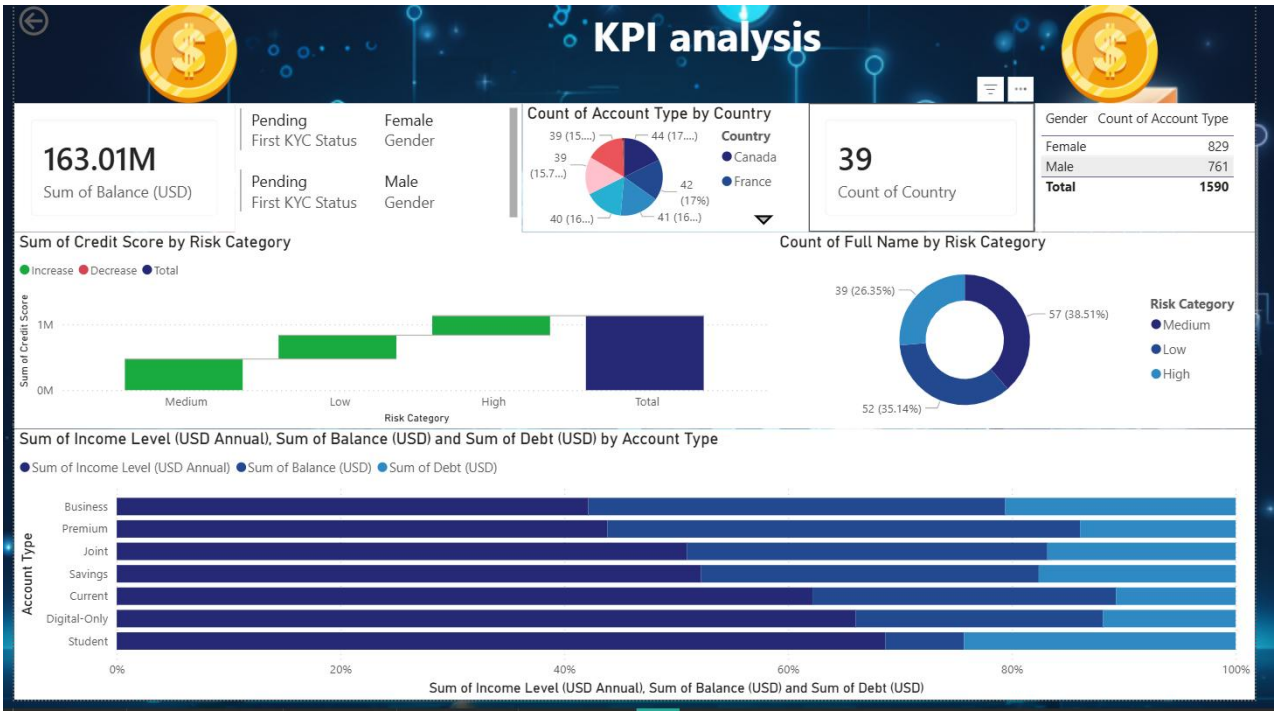
5.2 Project Links

Share the relevant links to your project work for evaluation.

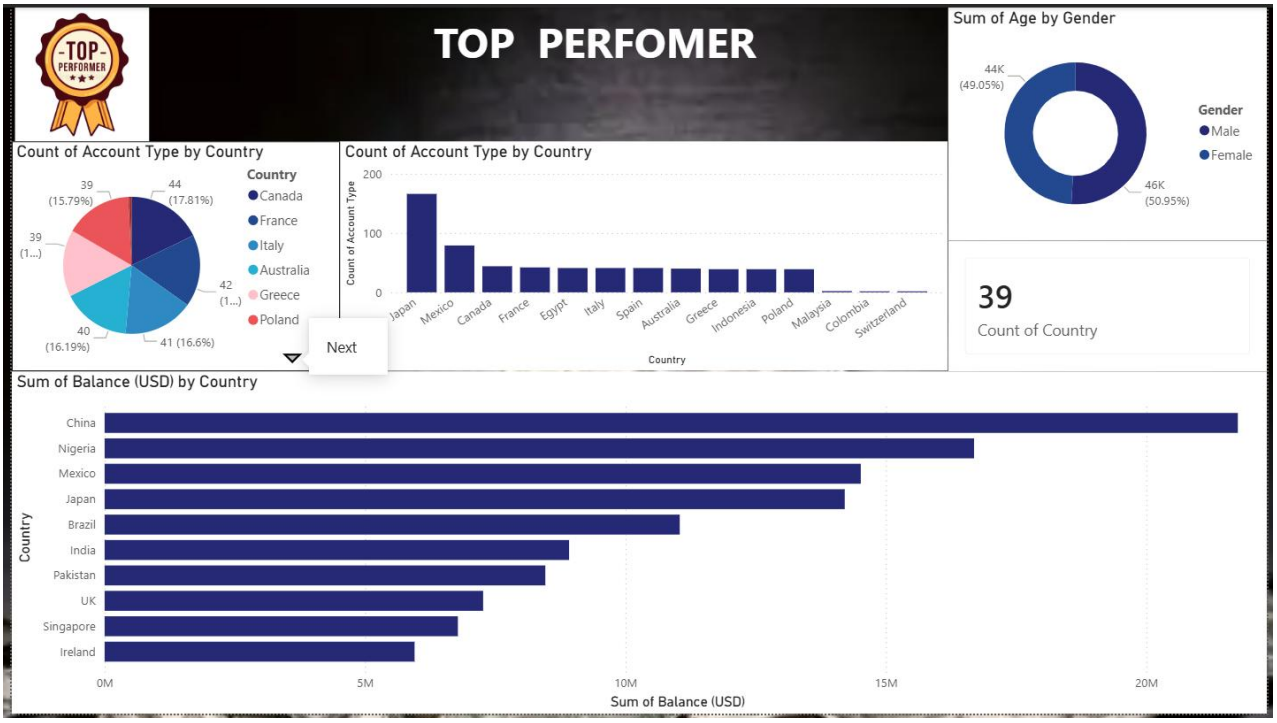
- GitHub Link of the Project: banking-customer-data-analysisbanking-customer-data-analysis
- Power BI
    1. Demographictrend



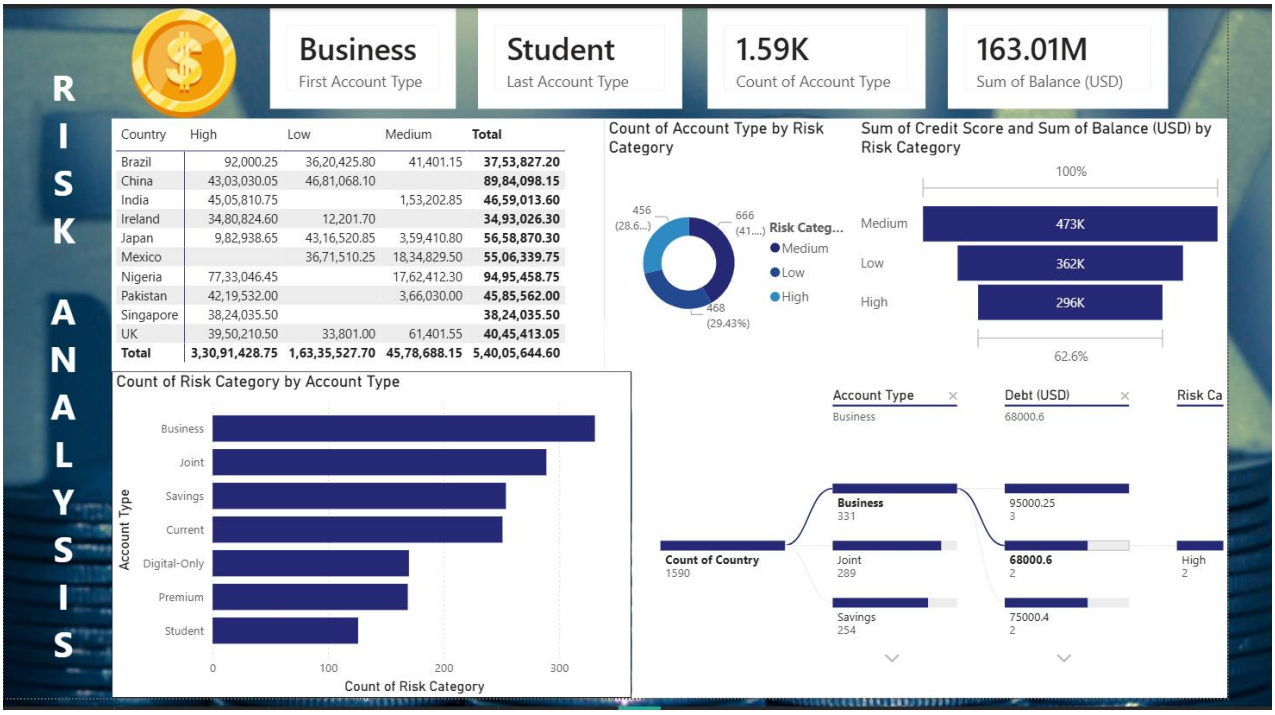
    2. KPI analysis



3. top perfomer



4. Risk category



Country	High	Low	Medium	Total
Brazil	92,000.25	36,20,425.80	41,401.15	**37,53,827.20**
China	43,03,030.05	46,81,068.10		**89,84,098.15**
India	45,05,810.75		1,53,202.85	**46,59,013.60**
Ireland	34,80,824.60	12,201.70		**34,93,026.30**
Japan	9,82,938.65	43,16,520.85	3,59,410.80	**56,58,870.30**
Mexico		36,71,510.25	18,34,829.50	**55,06,339.75**
Nigeria	77,33,046.45		17,62,412.30	**94,95,458.75**
Pakistan	42,19,532.00		3,66,030.00	**45,85,562.00**
Singapore	38,24,035.50			**38,24,035.50**
UK	39,50,210.50	33,801.00	61,401.55	**40,45,413.05**
Total	**3,30,91,428.75**	**1,63,35,527.70**	**45,78,688.15**	**5,40,05,644.60**

5.3 Limitations

Discuss the limitations of the current project or approach honestly.

- Limitation 1: The project was tested on a sample dataset form Kaggle and may not generalize to all real-world scenarios.
- Limitation 2: Microsoft Copilot access was limited during the project,
- Limitation 3: The dashboard does not support real-time data refresh due to tool limitations that is map tools in Power BI
- Limitation 4: Advanced features may required paid subscription in power BI.
- Limitation 5: Limited access to real time banking data in student level projects

5.4 Future Work

The suggestions for improving or extending the project in the future.

- Integrate real-time data pipelines using Power Automate or API connections.
- Apply machine learning models (e.g., forecasting, anomaly detection) on top of the cleaned dataset.
- Build a web-based interface or Power Apps solution for non-technical end users.
- Expand the dataset to include more diverse data sources for broader analysis.
- Develop a standardized prompt library for Copilot use in this domain.

5.5 Conclusion

This project focused on using Power BI and Microsoft Copilot to analyze and visualize banking data in a simple and effective way. The main objective was to make complex data easy to understand and support better decision-making. Through this project, interactive dashboards were created to present key information like transactions, customer behavior, and performance trends.

The use of Copilot helped in generating quick insights and summaries, which reduced manual effort and saved time. The project successfully showed how combining data visualization with AI can improve efficiency, accuracy, and productivity in banking analysis. Most of the objectives of the project were achieved, including data analysis, visualization, and AI-based insight generation.

Overall, this project added value by demonstrating a practical and user-friendly approach to banking analytics. It also helped in improving my knowledge and skills in data analysis, Power BI, and AI tools, making it a valuable learning experience.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[5] Reserch paper 1.)The influence of the practices of big data analytics applications on bank performance: filed study | VINE Journal of Information and Knowledge Management Systems | Emerald Publishing

[6] Reserch paper 2.)Analyzing Banking Data Using Business Intelligence: A Data Mining Approach | Springer Nature Link

[7] kaggle data set https://www.kaggle.com/datasets/realzippo/global-banking-customer-analytics-
dataset

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

```
= Table.RemoveColumns(#"Changed Type",{"Username", "Email", "Phone Number"})
```

	Full Name	Account Type	Country	City	Age	Gender
1	Carlos M. Rivera Jr.	Premium	US	New York	42	Male
2	Aiko Tanaka	Digital-Only	Japan	Tokyo	28	Female
3	Md. Rahim Uddin	Business	India	Mumbai	35	Male
4	Anna-Louise Clarke	Joint	UK	London	31	Female
5	John 'Jack' Thompson	Savings	US	Chicago	45	Male
6	Wei Zhang	Current	China	Shanghai	26	Male
7	Maria Silva Santos	Savings	Brazil	São Paulo	39	Female
8	Chiamaka Okoro	Digital-Only	Nigeria	Lagos	24	Female
9	Hans Müller	Business	Germany	Berlin	52	Male
10	Fatima Al Mansouri	Joint	UAE	Dubai	36	Female
11	James Wilson-Smith	Student	Canada	Toronto	21	Male
12	Yuki Nakamura	Premium	Japan	Osaka	48	Female
13	Rajesh Kumar Patel	Savings	India	Delhi	33	Male
14	Sarah Johnson-Williams	Current	UK	Manchester	29	Female
15	Mohammed Hassan	Business	UAE	Abu Dhabi	44	Male
16	Luciana Costa	Digital-Only	Brazil	Rio de Janeiro	27	Female
17	Kenji Sato	Premium	Japan	Kyoto	55	Male
18	Adebola Okafor	Savings	Nigeria	Abuja	38	Male
19	Emma-Rose Gallagher	Student	US	Los Angeles	20	Female
20	Chen Wei	Current	China	Beijing	32	Male
21	Sophie Schmidt	Savings	Germany	Hamburg	41	Female
22	David Chen-Li	Joint	Canada	Vancouver	37	Male
23	Priya Sharma	Digital-Only	India	Bangalore	26	Female
24	Ahmed Abdullah	Business	UAE	Sharjah	49	Male
25	Isabella Rossi	Current	Brazil	São Paulo	30	Female
26	Takashi Yamamoto	Premium	Japan	Tokyo	53	Male

Appendix B — Key Excel Formulas / DAX Measures Used

List any important formulas, functions, or DAX measures used in the project:

- =IFERROR(VLOOKUP(...), "Not Found") — used for data validation
- =COUNTIF(range, criteria) — used to count duplicates
- Power BI DAX:Total Amt.=SUM(Transaction[Amount])

Appendix C — AI Prompts Used

The key prompts that I used with Copilot during the project:

- Prompt 1: "Summarize banking data and identify the top 5 performing countries."
- Prompt 2: “Analyze the data set and identify the financial health of customers based on balance, debt, income, and credit score. Classify them into strong, moderate, and weak financial categories.”

- Prompt 3: "Analyze the dataset to identify high-risk customers and fraud patterns using risk category, credit score, and fraud flag."
- Prompt 4:”Analyze customer demographics such as age, gender, and country to identify key patterns and distributions.”